Algebra 6 — Proportion with a constant k · Laws of indices

Calculator allowed except in question 6. Marks in brackets. ★ = harder question.

1 y is directly proportional to x. When $x = 8$, $y = 12$. [3]

a Find a formula for y in terms of x. **b** Find y when $x = 10$.

c Find x when $y = 4.5$.

2 y is directly proportional to the square of x. When $x = 3$, $y = 45$. [4]

a Find a formula for y in terms of x. **b** Find y when $x = 4$.

c Find the positive value of x when $y = 125$.

3 y is directly proportional to $\sqrt{x}$. When $x = 9$, $y = 6$. [4]

a Find a formula for y in terms of x. **b** Find y when $x = 25$.

c Find x when $y = 1$.

4 The brightness I of a lamp, measured at a distance d metres from it, is inversely proportional to the square of the distance. At 2 m the brightness is 36 units. [6]

a Find a formula for I in terms of d. **b** Find the brightness at 6 m.

c At what distance is the brightness 9 units?

d The distance is doubled. What happens to the brightness? Explain.

5 A fixed amount of gas is squeezed in a cylinder. Its pressure P (in kPa) is inversely proportional to its volume V (in cm³). When $V = 240$, $P = 125$. [4]

a Find a formula for P in terms of V. **b** Find P when $V = 150$.

c Find V when $P = 400$.

6 Simplify, without a calculator. [4]

a $2a^{-3} \times 5a^{5}$ **b** $\left(c^{-6}\right)^{\frac{1}{3}}$ **c** $d^{\frac{3}{2}} \times \left(d^{\frac{1}{2}}\right)^{-3} \div d^{-1}$

d Find x if $\left(\frac{27}{8}\right)^{x} = \frac{4}{9}$.

Total: 25 marks

Check: "y varies as x^2" means $y = kx^2$ — always find k first from the given pair · inverse: $y = \frac{k}{x}, \frac{k}{x^2}, \frac{k}{\sqrt{x}}$ · multiply → add indices, divide → subtract, power of a power → multiply · $a^0 = 1$, $a^{-n} = \frac{1}{a^n}$, $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$.